

What is Pharmacometrics

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What is Pharmacometrics?

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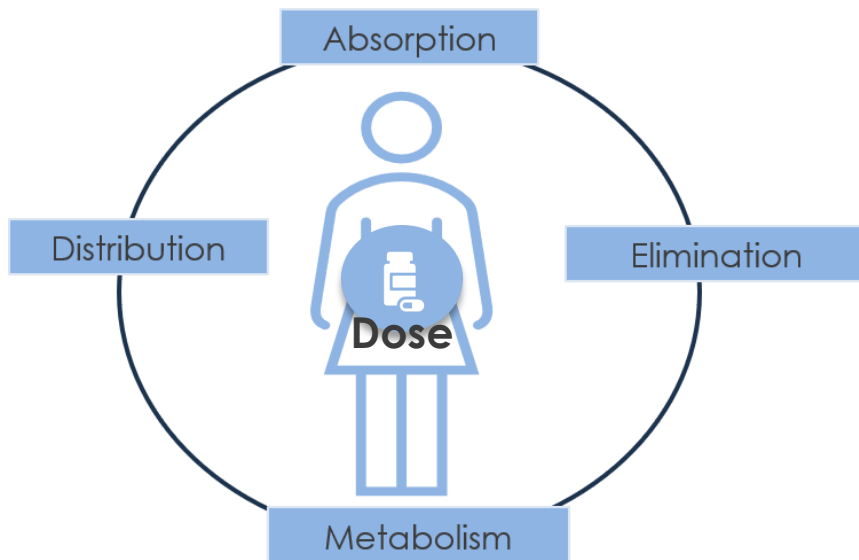
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What is a pharmacometric model

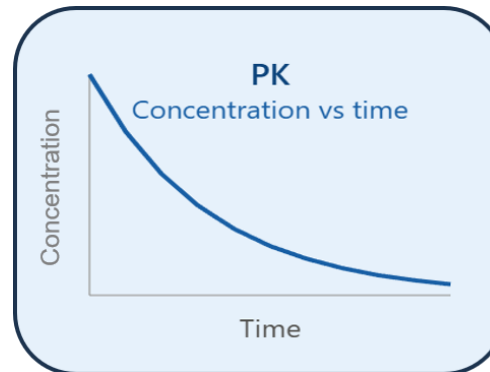
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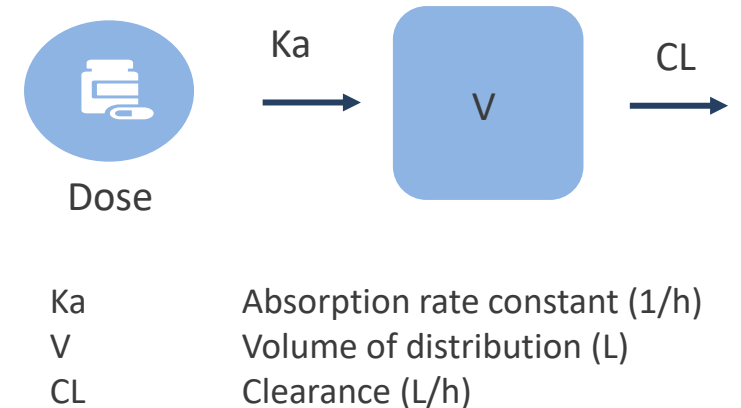
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PK data

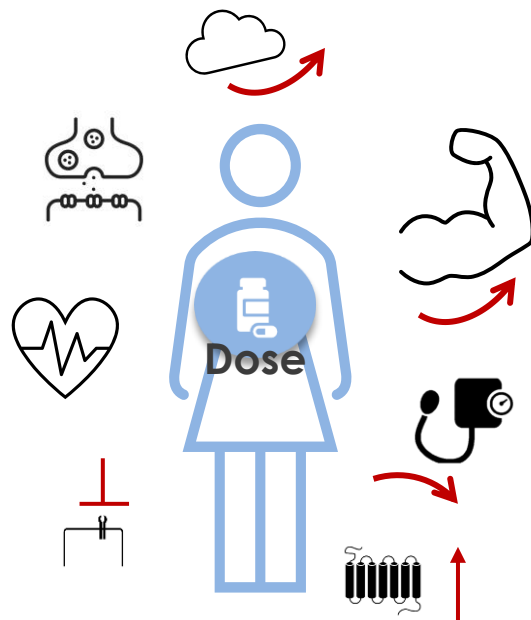


PK model

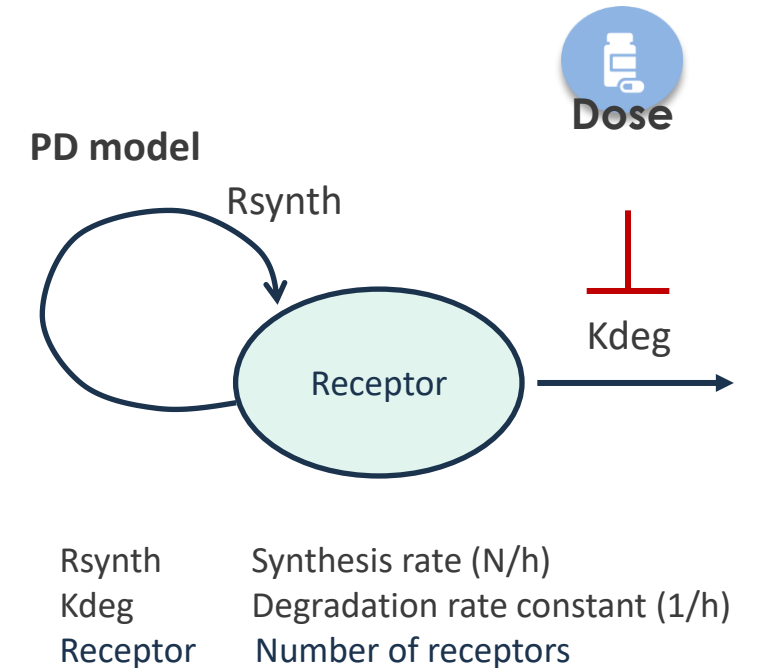
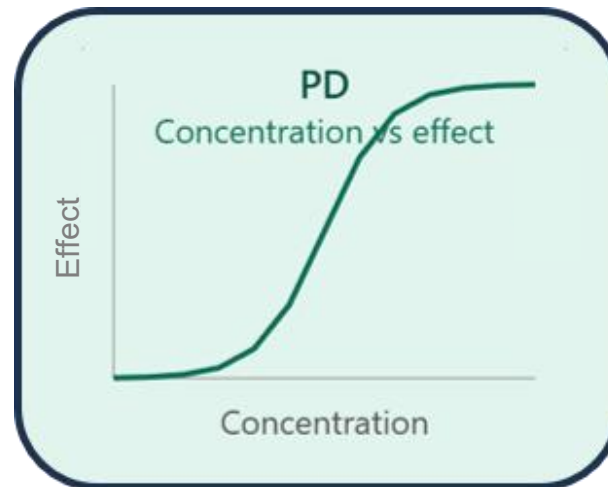


What is a pharmacometric model

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 - Pharmacokinetics (PK) or “*What the body does to the drug*”.
 - Pharmacodynamics (PD) or “*What the drug does to the body*”.

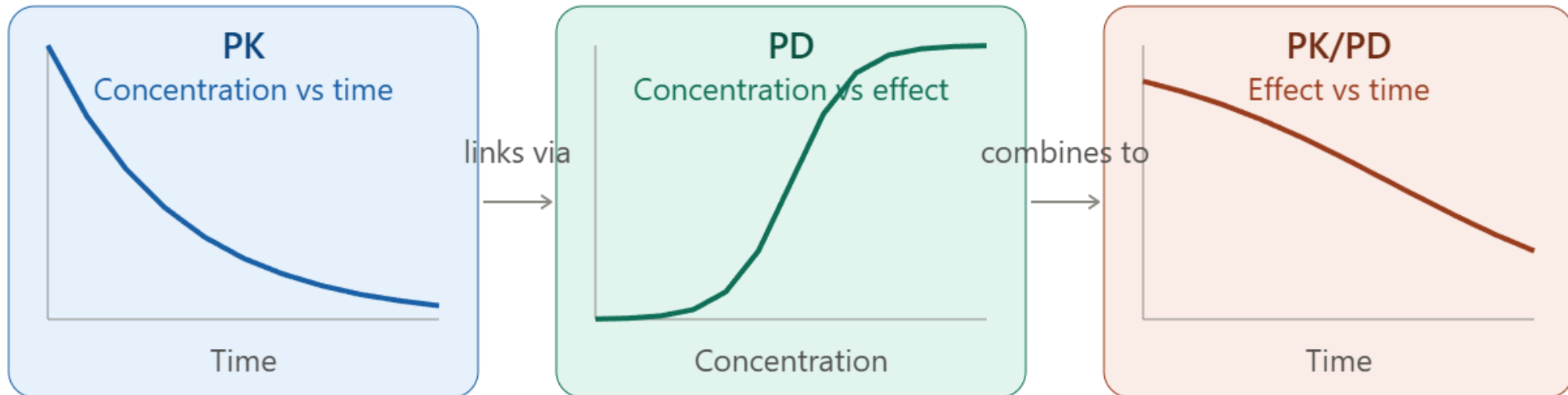


PD data



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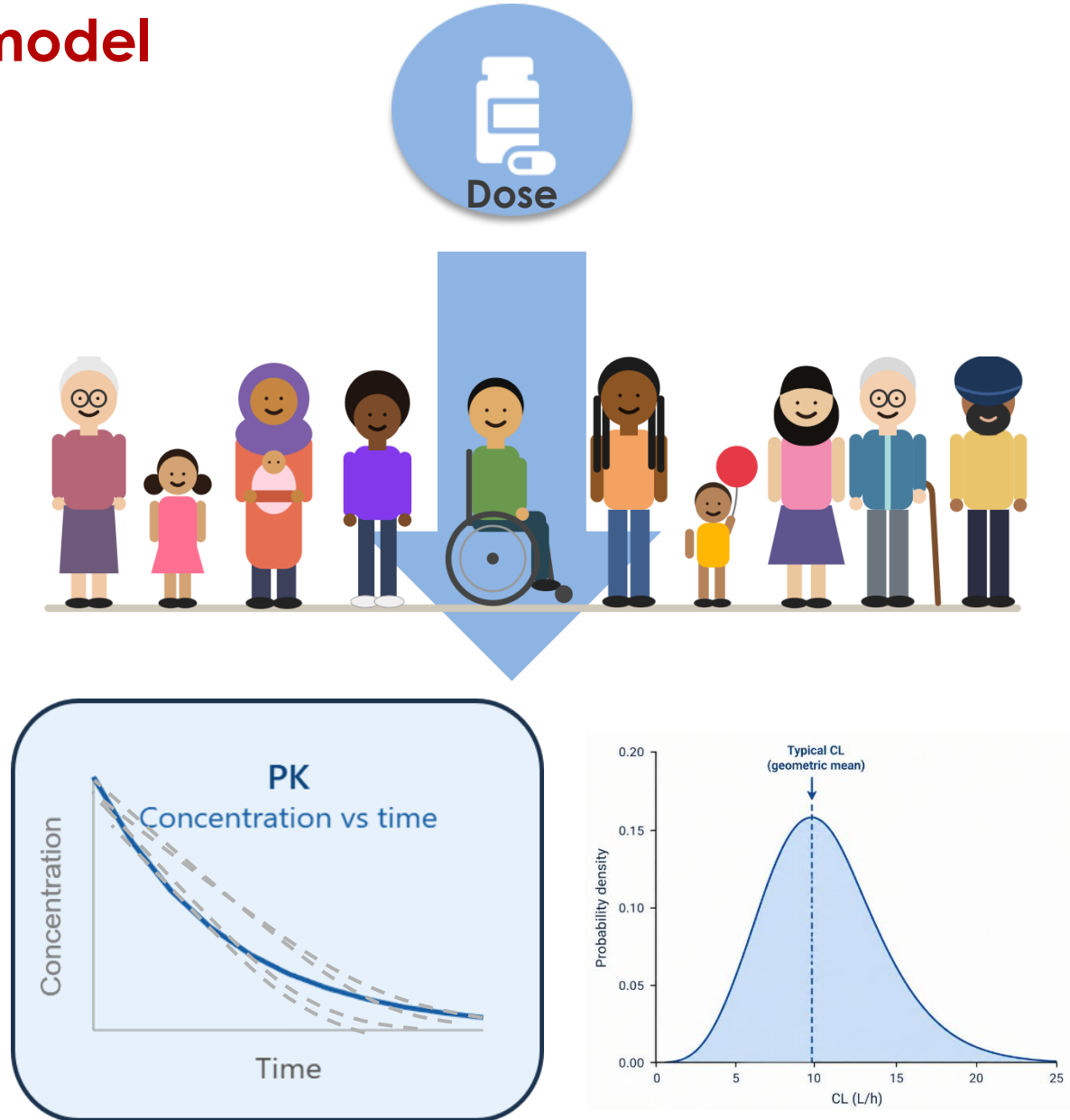
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- A pharmacometric model can help to understand how drugs behave in **different people**.
 - **Population model (Mixed-effects models)**

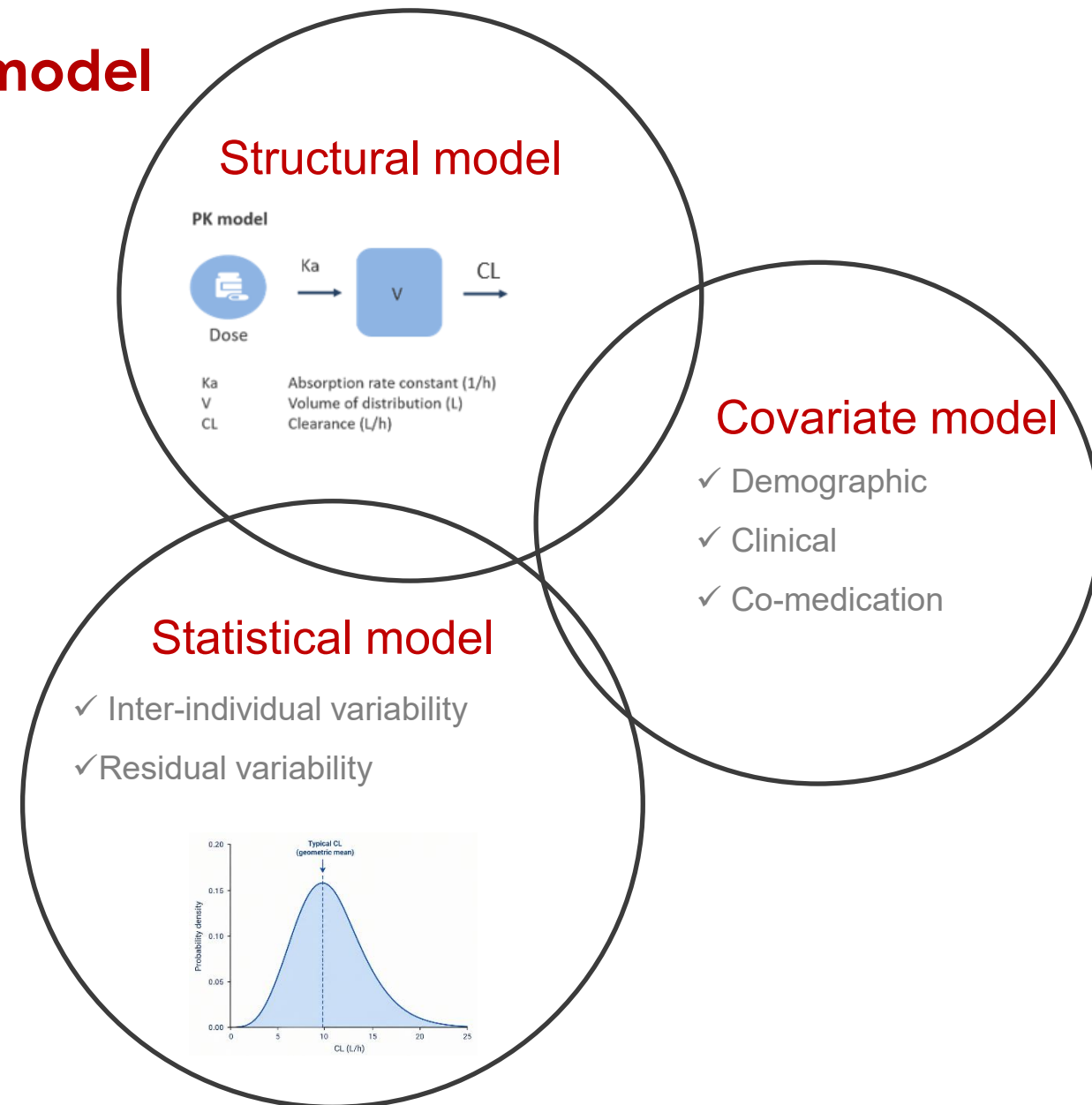
What is a population (mixed-effects) model

- A pharmacometric model can help to understand how drugs behave in different people.
- Giving the same dose to patients can result in differences in the concentrations but also in the drug response.
- Differences in drug response can be
 - **Predictable** (due to differences in body weight or kidney function, for instance) or
 - **Unpredictable** (differences between and within subjects that seem random, but likely reflect a lack of knowledge or data).
 - **Residual unexplained** variability is also dealt with (tackling things unconnected to individuals, such as variability in laboratory assays).



What is a population (mixed-effects) model

- **Structural model** – “What is the typical drug behavior?”
 - is the mathematical model (often ODEs) describing the typical behaviour within the population (Fixed Effects)
- **Statistical model** describes the differences within the population but also within a subjects.
 - “Why are patients different?” - Unpredictable (differences between and within subjects that seem random, but likely reflect a lack of knowledge or data)
 - “Why doesn’t the model fit every observation?” - Residual unexplained variability is also dealt with (tackling things unconnected to individuals, such as variability in laboratory assays).
- **Covariate model** - “Why do patients differ systematically?”
 - Predictable (due to differences in body weight or kidney function, for instance)



Why is it useful?

- **Questions pharmacometrics helps answer**
 - What dose should we use in our Phase 2 trial?
 - How does renal impairment affect drug exposure?
 - What is the probability of achieving target exposure in pediatric patients?
 - Can we predict efficacy from our Phase 2 data?
 - How should we adjust dosing in special populations?